IGCSE 0607 Practice Questions — E8.3 — Combined Questions & Markscheme

Magnitude of a vector

Total Questions: 15 — **Total Marks:** 100 Marks

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1 Vector $\mathbf{a} = \begin{pmatrix} -12 \\ 5 \end{pmatrix}$.

Calculate $|\mathbf{a}|$.

$|\mathbf{a}| =$ ..... **[2]**

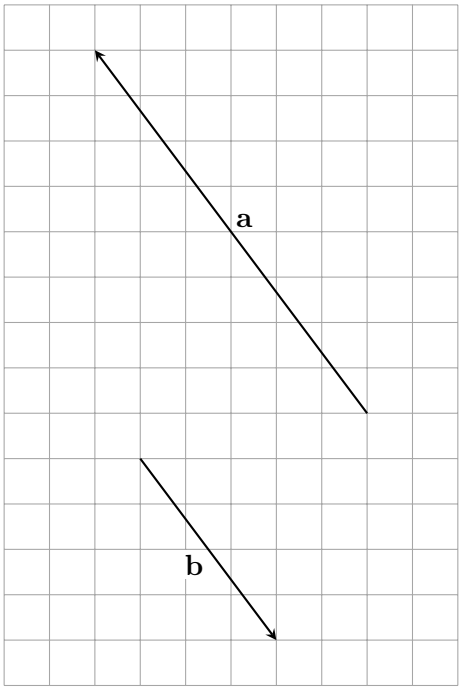
2 Vector $\mathbf{v}$ is given as

$$\mathbf{v} = \begin{pmatrix} 5 \\ -12 \end{pmatrix}$$

Work out $|\mathbf{v}|$, the magnitude of vector $\mathbf{v}$.

..... **[2]**

3 The diagram shows vectors **a** and **b** drawn on a square grid.



(a) Work out $|\mathbf{a}|$.

$|\mathbf{a}| =$ ..... **[2]**

(b) Work out $|2\mathbf{a} + 3\mathbf{b}|$.

$|2\mathbf{a} + 3\mathbf{b}| =$ ..... **[3]**

4 Given the vectors $\mathbf{p} = \begin{pmatrix} -8 \\ 15 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$.

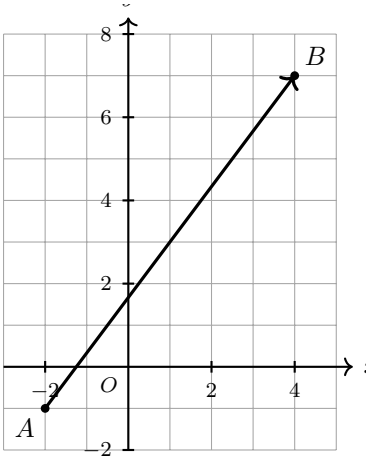
(a) Calculate $|\mathbf{p}|$.

..... **[2]**

(b) Calculate $|\mathbf{p} + 2\mathbf{q}|$.

..... **[3]**

5 The diagram shows points $A(-2,-1)$ and $B(4,7)$ on a coordinate grid.



(a) Write down $\vec{AB}$ as a column vector.

$\vec{AB} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

(b) Calculate $|\vec{AB}|$.

$|\vec{AB}| =$ ..... units **[2]**

(c) The vector $\mathbf{v}$ is given by $\mathbf{v} = \begin{pmatrix} -12 \\ -5 \end{pmatrix}$.

Calculate

(i) $|\mathbf{v}|$,

$|\mathbf{v}| =$ ..... units **[2]**

(ii) $|\vec{AB} + \mathbf{v}|$, giving your answer in the form $k\sqrt{5}$, where k is an integer.

$|\vec{AB} + \mathbf{v}| =$ ..... units **[2]**

6 **(a)** The vector $\mathbf{a} = \begin{pmatrix} p \\ p \end{pmatrix}$, where $p > 0$.

Given that $|\mathbf{a}| = \sqrt{18}$, find the value of p.

$p =$ ..... **[2]**

(b) The vectors $\mathbf{u}$ and $\mathbf{v}$ are given by

$$\mathbf{u} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \qquad \mathbf{v} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

A vector $\mathbf{w}$ is defined as $\mathbf{w} = \mathbf{u} + t\mathbf{v}$.

Given that $|\mathbf{w}| = \sqrt{29}$, find the two possible values of t.

$t =$ ..... , or $t =$ ..... **[5]**

7 Vector $\mathbf{u} = \begin{pmatrix} x \\ x+2 \end{pmatrix}$.

(a) The magnitude of vector **u** is 10.
Show that $x^2 + 2x - 48 = 0$.

[3]

(b) Solve the equation

$$x^2 + 2x - 48 = 0$$

$x =$ ..... , $x =$ ..... **[2]**

(c) Hence write down the two possible column vectors for **u**.

[2]

8 Relative to an origin O, the position vectors of points A and B are $\vec{OA} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}$ and $\vec{OB} = \begin{pmatrix} -2 \\ 6 \end{pmatrix}$.

(a) Calculate the exact value of

(i) $|\vec{OA}|$, giving your answer in the form $a\sqrt{10}$, where a is an integer,

$|\vec{OA}| =$ ..... **[1]**

(ii) $|\vec{AB}|$, giving your answer in the form $b\sqrt{5}$, where b is an integer.

$|\vec{AB}| =$ ..... **[2]**

(b) Show that triangle OAB is both isosceles and right-angled.

[3]

(c) Point P lies on the line segment AB such that $AP : PB = 3 : 1$.

(i) Find $\vec{OP}$ as a column vector.

$\vec{OP} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(ii) Point Q has position vector $\vec{OQ} = \begin{pmatrix} 8 \\ m \end{pmatrix}$, where $m < 0$.

Given that $|\vec{PQ}| = 17$, find the value of m.

$m =$ ..... **[3]**

9 Points P and Q have coordinates $(4, -2)$ and $(-1, 10)$ respectively.

(a) Find $\vec{PQ}$.

$\vec{PQ} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

(b) Calculate $|\vec{PQ}|$.

$|\vec{PQ}| =$ ..... **[2]**

(c) Vector $\mathbf{m} = \begin{pmatrix} 3 \\ c \end{pmatrix}$.

Given that $|\mathbf{m}| = |\vec{PQ}|$, find the possible values of c.

$c =$ ..... , $c =$ ..... **[3]**

10 Points A, B, and C have coordinates $(3, -2)$, $(-5, y)$, and $(7, 1)$ respectively.

(a) Calculate $|\vec{AC}|$.

..... **[2]**

(b) The magnitude of vector $\vec{AB}$ is 17.
Work out the two possible values of y.

$y =$ ..... , $y =$ ..... **[5]**

11 The points P, Q and R have coordinates $(2, -3)$, $(k, 5)$ and $(4, 1)$ respectively.

(a) Given that $|\vec{PQ}| = 10$, find the two possible values of k.

$k =$ ..... , or $k =$ ..... **[4]**

(b) Using the positive value of k from part (a), calculate the magnitude of $\vec{QR}$.
Give your answer correct to 3 significant figures.

$|\vec{QR}| =$ ..... **[3]**

12 **(a)** The points A and B have coordinates $(2, -3)$ and $(9, 4)$ respectively.
Work out the column vector $\vec{AB}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) Calculate $|\vec{AB}|$.
Give your answer correct to 3 significant figures.

..... **[2]**

(c) Vector $\mathbf{p} = \begin{pmatrix} k \\ 4 \end{pmatrix}$.
Given that $|\mathbf{p}| = 5$, find the two possible values of k.

$k =$ ..... , $k =$ ..... **[3]**

13 $\mathbf{p} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} -1 \\ 5 \end{pmatrix}$.

(a) Calculate $|2\mathbf{p} - \mathbf{q}|$, giving your answer correct to 3 significant figures.

$|2\mathbf{p} - \mathbf{q}| =$ ..... **[3]**

(b) The vector $\mathbf{r} = k\mathbf{p}$, where $k < 0$ and $|\mathbf{r}| = 35$.
Find the column vector $\mathbf{r}$.

$\mathbf{r} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[3]**

(c) The vector $\mathbf{w} = \begin{pmatrix} t+3 \\ 2t-1 \end{pmatrix}$ has magnitude $\sqrt{34}$.
Find the two possible values of t.

$t =$ ..... , $t =$ ..... **[4]**

14 Vector $\mathbf{v} = \begin{pmatrix} k+3 \\ 2k-1 \end{pmatrix}$, where k is a constant.

The magnitude of $\mathbf{v}$ is $\sqrt{98}$.

(a) Show that $5k^2 + 2k - 88 = 0$.

[3]

(b) Solve the equation $5k^2 + 2k - 88 = 0$ to find the possible values of k.

$k =$ ..... , $k =$ ..... **[3]**

(c) Given that $k > 0$, write down the column vector $\mathbf{v}$.

$\mathbf{v} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[1]**

15 Given the vectors $\mathbf{u} = \begin{pmatrix} 2k \\ k-3 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} k+1 \\ 2 \end{pmatrix}$.

(a) Given that $|\mathbf{u}+\mathbf{v}| = \sqrt{50}$, show that $5k^2 + 2k - 24 = 0$.

[4]

(b) Solve the equation

$$5k^2 + 2k - 24 = 0$$

$k =$ ..... , $k =$ ..... **[4]**

(c) Hence, find the two possible column vectors for $\mathbf{u}$.

$\mathbf{u} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ ..... , $\mathbf{u} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ ..... **[2]**

1 $\sqrt{(-12)^2+5^2}$ or $\sqrt{144+25}$ **[M1]**

13 **[A1]**

[Back to Question] *[Total Marks: 2]*

2 $\sqrt{5^2+(-12)^2}$ or $\sqrt{25+144}$ **[M1]**

13 **[A1]**

[Back to Question] *[Total Marks: 2]*

3 **(a)** Determines column vector $\mathbf{a} = \begin{pmatrix}-6\\8\end{pmatrix}$ **[M1]**

10 **[A1]**

(b) Determines column vector $\mathbf{b} = \begin{pmatrix}3\\-4\end{pmatrix}$ **[B1]**

Calculates resultant vector $2\begin{pmatrix}-6\\8\end{pmatrix}+3\begin{pmatrix}3\\-4\end{pmatrix}=\begin{pmatrix}-12\\16\end{pmatrix}+\begin{pmatrix}9\\-12\end{pmatrix}=\begin{pmatrix}-3\\4\end{pmatrix}$ **[M1]**

5 **[A1]**

[Back to Question] *[Total Marks: 5]*

4 **(a)** $\sqrt{(-8)^2+15^2}$ or $\sqrt{64+225}$ **[M1]**

17 **[A1]**

(b) $\mathbf{p}+2\mathbf{q} = \begin{pmatrix}-8\\15\end{pmatrix}+\begin{pmatrix}4\\-8\end{pmatrix}=\begin{pmatrix}-4\\7\end{pmatrix}$ **[M1]**

$\sqrt{(-4)^2+7^2}$ or $\sqrt{16+49}$ **[M1]**

$\sqrt{65}$ **[A1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** $\begin{pmatrix}6\\8\end{pmatrix}$ **[B1]**

(b) $|\vec{AB}| = \sqrt{6^2+8^2} = \sqrt{36+64}$ **[M1]**

10 **[A1]**

(c) **(i)** $|\mathbf{v}| = \sqrt{(-12)^2+(-5)^2} = \sqrt{144+25}$ **[M1]**

13 **[A1]**

(ii) $\vec{AB}+\mathbf{v} = \begin{pmatrix}6\\8\end{pmatrix}+\begin{pmatrix}-12\\-5\end{pmatrix}=\begin{pmatrix}-6\\3\end{pmatrix}$ **[M1]**

$\sqrt{(-6)^2+3^2} = \sqrt{45} = 3\sqrt{5}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

6 **(a)** Sets up equation: $p^2 + p^2 = 18$ **[M1]**

3 **[A1]**

(b) Finds expression for **w**: $\begin{pmatrix} 4-t \\ 1+2t \end{pmatrix}$ **[M1]**

Forms magnitude squared equation: $(4-t)^2 + (1+2t)^2 = 29$ **[M1]**

Expands brackets correctly: $16 - 8t + t^2 + 1 + 4t + 4t^2 = 29$ **[M1]**

Rearranges to a 3-term quadratic: $5t^2 - 4t - 12 = 0$ **[M1]**

$t = 2$, or $t = -1.2$ (or $-\frac{6}{5}$) **[A1]**

[Back to Question] *[Total Marks: 7]*

7 **(a)** $\sqrt{x^2 + (x+2)^2} = 10$ or $x^2 + (x+2)^2 = 10^2$ **[M1]**

$x^2 + x^2 + 4x + 4 = 100$ **[M1]**

$2x^2 + 4x - 96 = 0$ leading to $x^2 + 2x - 48 = 0$ with no errors seen **[A1]**

(b) $(x+8)(x-6) = 0$ or correct substitution into quadratic formula **[M1]**

$x = -8, x = 6$ **[A1]**

(c) $\mathbf{u} = \begin{pmatrix} -8 \\ -6 \end{pmatrix}$ **[B1]**

$\mathbf{u} = \begin{pmatrix} 6 \\ 8 \end{pmatrix}$ **[B1]**

[Back to Question] *[Total Marks: 7]*

8 **(a)** **(i)** $\sqrt{6^2 + 2^2} = \sqrt{40} = 2\sqrt{10}$ **[B1]**

(ii) $\vec{AB} = \begin{pmatrix} -2 \\ 6 \end{pmatrix} - \begin{pmatrix} 6 \\ 2 \end{pmatrix} = \begin{pmatrix} -8 \\ 4 \end{pmatrix}$ **[M1]**

$|\vec{AB}| = \sqrt{(-8)^2 + 4^2} = \sqrt{80} = 4\sqrt{5}$ **[A1]**

(b) $|\vec{OB}| = \sqrt{(-2)^2 + 6^2} = \sqrt{40} = 2\sqrt{10}$ **[M1]**

$|\vec{OA}| = |\vec{OB}| = 2\sqrt{10}$, so triangle OAB is isosceles **[A1]**

$|\vec{OA}|^2 + |\vec{OB}|^2 = 40 + 40 = 80 = |\vec{AB}|^2$, so angle $AOB = 90°$ **[A1]**

(c) **(i)** $\vec{OP} = \vec{OA} + \frac{3}{4}\vec{AB} = \begin{pmatrix} 6 \\ 2 \end{pmatrix} + \frac{3}{4}\begin{pmatrix} -8 \\ 4 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 0 \\ 5 \end{pmatrix}$ **[A1]**

(ii) $\vec{PQ} = \begin{pmatrix} 8 \\ m \end{pmatrix} - \begin{pmatrix} 0 \\ 5 \end{pmatrix} = \begin{pmatrix} 8 \\ m-5 \end{pmatrix}$ **[M1]**

$8^2 + (m-5)^2 = 17^2$ giving $(m-5)^2 = 225$ **[M1]**

$m - 5 = -15$ (since $m < 0$), so $m = -10$ **[A1]**

[Back to Question] *[Total Marks: 11]*

9 **(a)** $\begin{pmatrix} -5 \\ 12 \end{pmatrix}$ **[B1]**

(b) $\sqrt{(-5)^2 + 12^2}$ or $\sqrt{25 + 144}$ **[M1]**

13 **[A1]**

(c) $3^2 + c^2 = 13^2$ or $9 + c^2 = 169$ **[M1]**

$c^2 = 160$ **[M1]**

12.6 and -12.6 (or $\pm 4\sqrt{10}$) **[A1]**

[Back to Question] *[Total Marks: 6]*

10 **(a)** $\vec{AC} = \begin{pmatrix} 7 - 3 \\ 1 - (-2) \end{pmatrix}$ or $\begin{pmatrix} 4 \\ 3 \end{pmatrix}$ **[M1]**

5 **[A1]**

(b) $\vec{AB} = \begin{pmatrix} -8 \\ y + 2 \end{pmatrix}$ **[B1]**

$\sqrt{(-8)^2 + (y + 2)^2} = 17$ **[M1]**

$64 + (y + 2)^2 = 289$ or $(y + 2)^2 = 225$ **[M1]**

$y + 2 = 15$ or $y + 2 = -15$ **[M1]**

$y = 13$, $y = -17$ **[A1]**

[Back to Question] *[Total Marks: 7]*

11 **(a)** Expresses vector $\vec{PQ}$ as $\begin{pmatrix} k - 2 \\ 5 - (-3) \end{pmatrix} = \begin{pmatrix} k - 2 \\ 8 \end{pmatrix}$ **[M1]**

Sets up magnitude equation: $(k - 2)^2 + 8^2 = 10^2$ **[M1]**

Simplifies to solve for k: $(k - 2)^2 = 36$ **[M1]**

$k = 8$, or $k = -4$ **[A1]**

(b) Finds $\vec{QR}$ using $Q(8, 5)$ and $R(4, 1)$: $\vec{QR} = \begin{pmatrix} 4 - 8 \\ 1 - 5 \end{pmatrix} = \begin{pmatrix} -4 \\ -4 \end{pmatrix}$ **[M1]**

Calculates magnitude: $\sqrt{(-4)^2 + (-4)^2} = \sqrt{32}$ **[M1]**

5.66 **[A1]**

[Back to Question] *[Total Marks: 7]*

12 **(a)** $\begin{pmatrix} 9 - 2 \\ 4 - (-3) \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 7 \\ 7 \end{pmatrix}$ **[A1]**

(b) $\sqrt{7^2 + 7^2}$ or $\sqrt{98}$ **[M1]**

9.90 **[A1]**

(c) $\sqrt{k^2+4^2}=5$ or $k^2+4^2=5^2$ **[M1]**

$k^2=25-16$ or $k^2=9$ **[M1]**

$k=3,\ k=-3$ **[A1]**

[Back to Question] *[Total Marks: 7]*

13 **(a)** $2\mathbf{p}-\mathbf{q}=\begin{pmatrix}8\\-6\end{pmatrix}-\begin{pmatrix}-1\\5\end{pmatrix}=\begin{pmatrix}9\\-11\end{pmatrix}$ **[M1]**

$|2\mathbf{p}-\mathbf{q}|=\sqrt{9^2+(-11)^2}=\sqrt{202}$ **[M1]**

14.2 **[A1]**

(b) $|\mathbf{p}|=\sqrt{4^2+(-3)^2}=5$ **[B1]**

$k=-\dfrac{35}{5}=-7$ **[M1]**

$\begin{pmatrix}-28\\21\end{pmatrix}$ **[A1]**

(c) $(t+3)^2+(2t-1)^2=34$ **[M1]**

$t^2+6t+9+4t^2-4t+1=34$ simplifying to $5t^2+2t-24=0$ **[M1]**

$(5t+12)(t-2)=0$ **[M1]**

$t=2$ and $t=-2.4$ (or $-\dfrac{12}{5}$) **[A1]**

[Back to Question] *[Total Marks: 10]*

14 **(a)** $(k+3)^2+(2k-1)^2=98$ **[M1]**

$k^2+6k+9+4k^2-4k+1=98$ **[M1]**

$5k^2+2k+10=98$ leading to $5k^2+2k-88=0$ with no errors seen **[A1]**

(b) $(5k+22)(k-4)=0$ or $\dfrac{-2\pm\sqrt{2^2-4(5)(-88)}}{2(5)}$ **[M2]**

If M0, allow [M1] for a correct partial factorisation e.g. $(5k+a)(k+b)$ where $ab=-88$ or correct substitution into quadratic formula with one sign error

4 and -4.4 **[A1]**

(c) $\begin{pmatrix}7\\7\end{pmatrix}$ **[B1]**

[Back to Question] *[Total Marks: 7]*

15 **(a)** $\mathbf{u}+\mathbf{v}=\begin{pmatrix}3k+1\\k-1\end{pmatrix}$ **[M1]**

$(3k+1)^2+(k-1)^2=(\sqrt{50})^2$ **[M1]**

$9k^2+6k+1+k^2-2k+1=50$ **[M1]**

$10k^2+4k+2=50 \implies 5k^2+2k-24=0$ **[A1]**

(b) $(5k + a)(k + b) = 0$ where $ab = -24$ or $5b + a = 2$, or correct substitution into quadratic formula **[M1]**

$(5k + 12)(k - 2) = 0$ **[M1]**

$k = 2$ **[A1]**

$k = -2.4$ or $-\frac{12}{5}$ **[A1]**

(c) Substituting $k = 2$ and $k = -2.4$ into $\begin{pmatrix} 2k \\ k - 3 \end{pmatrix}$ **[M1]**

$\begin{pmatrix} 4 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} -4.8 \\ -5.4 \end{pmatrix}$ **[A1]**

[Back to Question] *[Total Marks: 10]*